



VACCINATION EVENTS ANALYSIS

From a raw event log to a governed star schema and a decision-ready KPI dashboard.

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25,983
EVENTS

13
MONTHS

5
SITES



FIVE PARTS

AGENDA

01	Dataset overview	What the vaccination event log contains and covers
02	Data model	Star schema: one fact table, three dimensions
03	Data quality findings	Three issues found, diagnosed and resolved
04	Primary KPI and dashboard	Vaccination volume by time, place and population
05	Insights and recommendations	What the numbers say and what to do next



ANONYMISED INDIVIDUAL EVENT LOG

DATASET OVERVIEW

Used to analyse service volume across time, location and population group.

COVERAGE WINDOW

1 Mar 2000 – 28 Mar 2001

13 months of service activity across 5 hospital sites, serving 6 provinces

25,983

VACCINATION EVENTS

5

HOSPITAL LOCATIONS

5

VACCINE PRODUCTS

GRAIN

1 row = 1 vaccination event

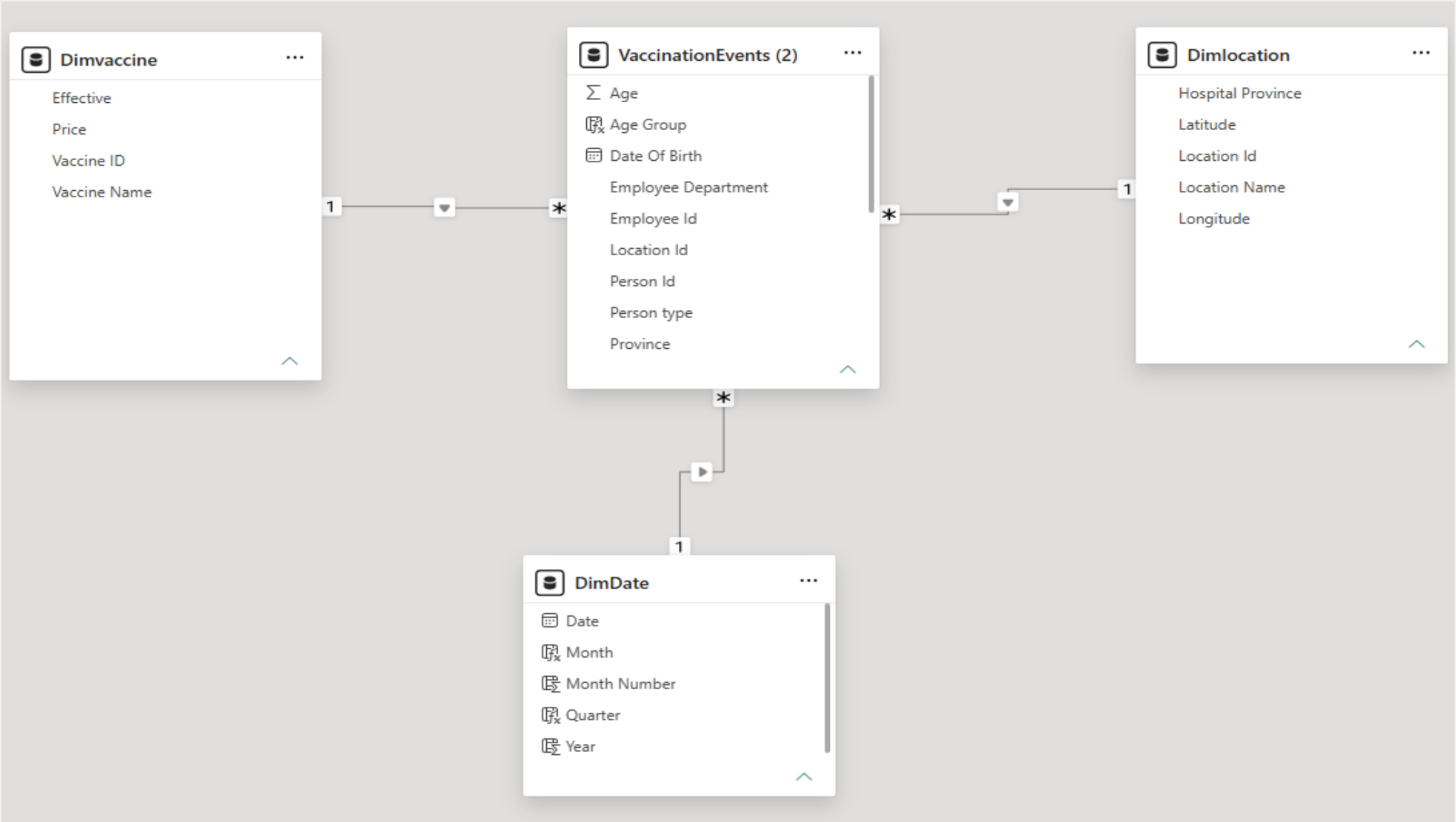
Keyed on Vaccination Id – 25,983 rows in the raw file, no duplicates

Person Id is not a reliable count of people: 37 Ids are shared by two people each – composite-key fix identified, not yet applied to the live model. See Data Quality Findings.



ONE FACT TABLE, THREE CONFORMED DIMENSIONS

DATA MODEL – STAR SCHEMA



Primary KPI = COUNT(Vaccination Id) on the fact table – independent of every dimension. Person attributes sit on the fact table, so distinct-person counting inherits the 37 known Id collisions.



DIAGNOSED WITH EVIDENCE, NOT ASSUMED

DATA QUALITY FINDINGS

LOW IMPACT

RESOLVED

Duplicate column

“Vaccine ID” and “Vaccine Id” matched on all 25,983 rows.

Fix applied: dropped the duplicate before modelling.

MEDIUM IMPACT

RESOLVED

Inconsistent province labels

Bangkok appears four ways — “Bangkok”, “กรุงเทพมหานคร”, “กรุงเทพฯ”, “กทม.” — the three non-standard spellings covering 4,561 rows, with “Bangkok Noi” and “Nakohn Pathom” adding 1,105 more: 5,666 rows of 25,983.

Fix applied: normalised to one standard label before aggregating by province.

HIGH IMPACT

OPEN

Person Id collisions

37 of 9,963 Person Ids are shared by two unrelated people — a narrow anonymised ID space colliding by chance, confirmed by self-consistent DOB, sex and age splits.

Recommended: composite Person Key (Person Id + Date Of Birth) → 10,000 correctly identified individuals, no records lost.



PERSON ID COLLISIONS — THE EVIDENCE

Every colliding Id splits into two internally consistent identities. Nothing is misspelled or mistyped — two different people were assigned the same anonymised number.

Person Id 3202930000000 — seven events, two people

EVENT	DATE OF BIRTH	SEX	AGE	VACCINE	PROVINCE
1000489	29/11/1947	Male	53	XinoV	Bangkok
1001328	29/11/1947	Male	53	XinoV	Bangkok
1003154	29/11/1947	Male	53	Moxer	Bangkok
1005149	29/11/1947	Male	53	Moxer	Bangkok
1008739	26/06/1900	Female	100	AstraX	Samut Prakan
1018864	26/06/1900	Female	100	AstraX	Samut Prakan
1023031	26/06/1900	Female	100	Moxer	Samut Prakan

Scale

37 Ids hold two dates of birth; 21 also split on sex and 26 span two provinces. All carry 4–7 events.

What it affects

Distinct-person and dose-sequence measures only. Total Vaccinations counts the fact table, so volume, dates, sites and vaccines are unaffected.

The fix

Composite Person Key = Person Id + Date Of Birth, added as a calculated column on the fact table. Resolves to 10,000 individuals with no records lost; widen the anonymisation range upstream so the workaround expires.



PRIMARY KPI

TOTAL VACCINATIONS

Total Vaccinations = COUNT(Vaccination Id)

Why this measure: the brief asks for vaccine service volume – how many doses were administered, not how many people were reached. Counting the fact table's grain (1 row = 1 event) answers that directly, with no joins required.

Secondary KPI: Distinct Persons = DISTINCTCOUNT(Person Key) – measures reach and coverage, and needs the composite Person Key before it can be trusted.

25,983

TOTAL VACCINATIONS

66.1

AVG. PER DAY

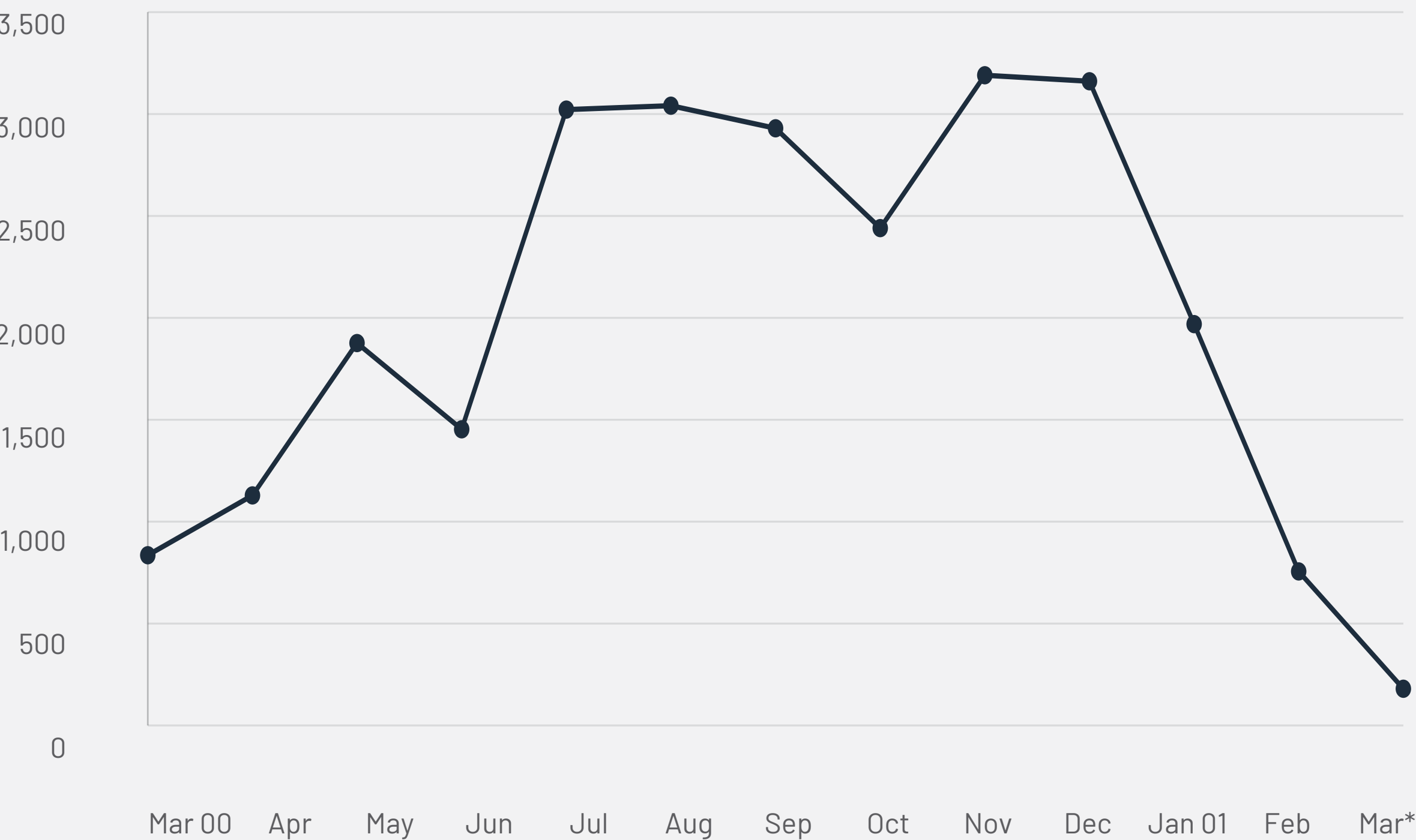
13

MONTHS OF ACTIVITY



TOTAL VACCINATIONS BY MONTH · MAR 2000 – MAR 2001

VOLUME OVER TIME



PEAK MONTH

Nov 2000

3,190 vaccinations

SUSTAINED PLATEAU

Jul – Dec 2000

~2,960 avg./month – 6 consecutive months above 2,400

TAPER

Jan – Mar 2001

Volume falls sharply

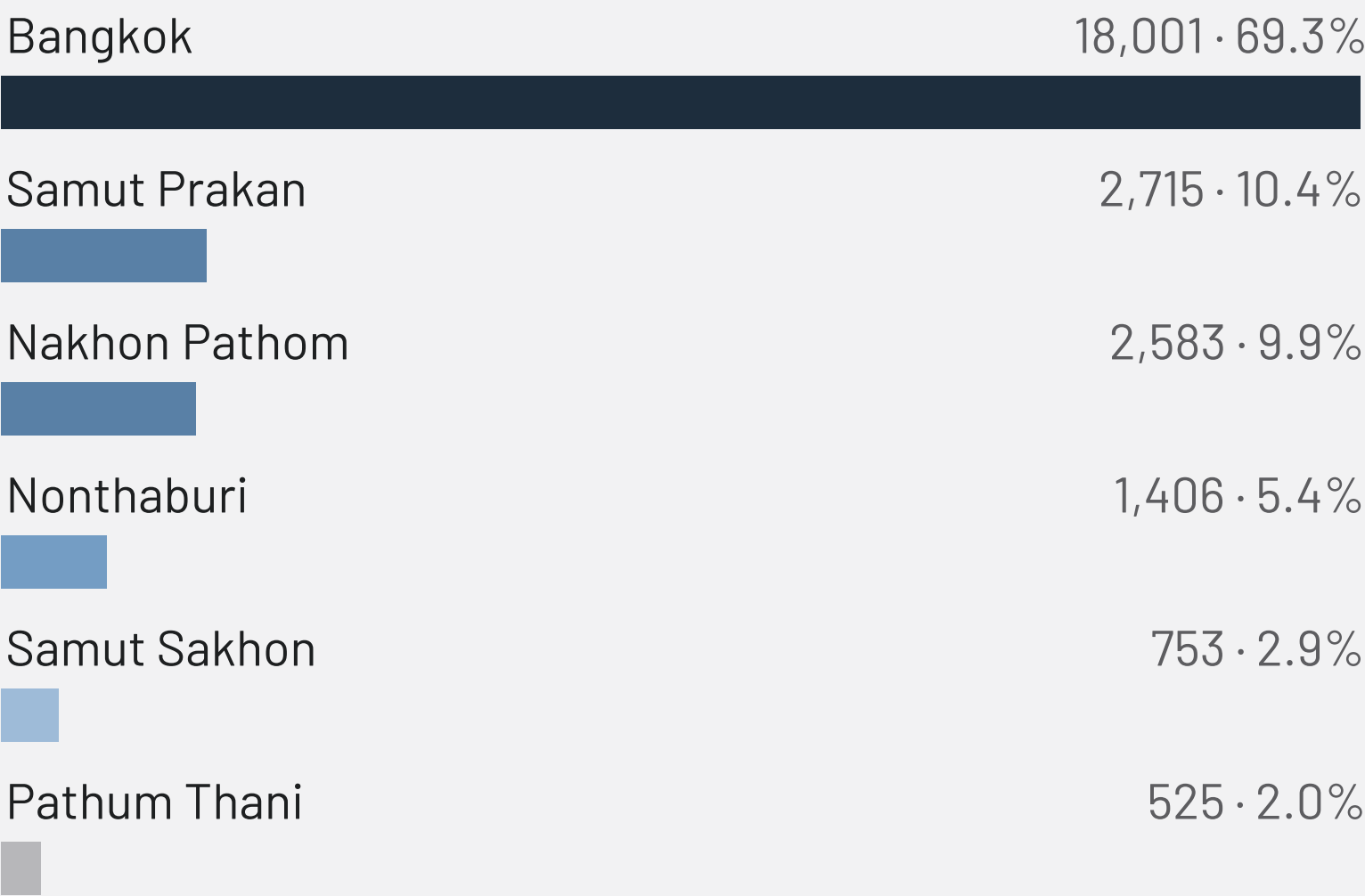


6 PATIENT PROVINCES · 5 HOSPITAL SITES

VOLUME BY LOCATION

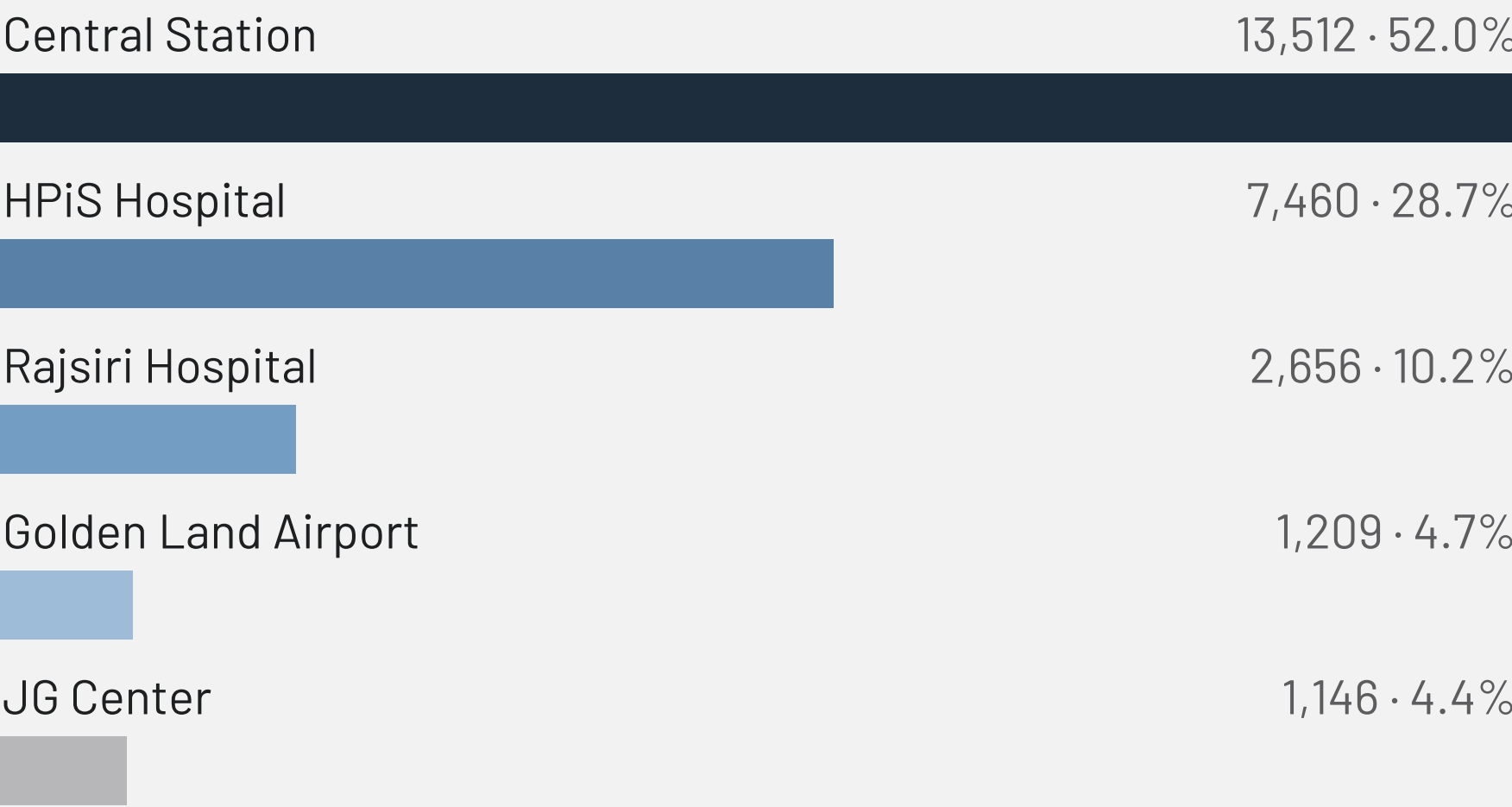
Service volume is heavily concentrated — by patient province and, within Bangkok, by site.

Patient province (normalised)



90.9% of doses are given at Bangkok sites.

Delivery site



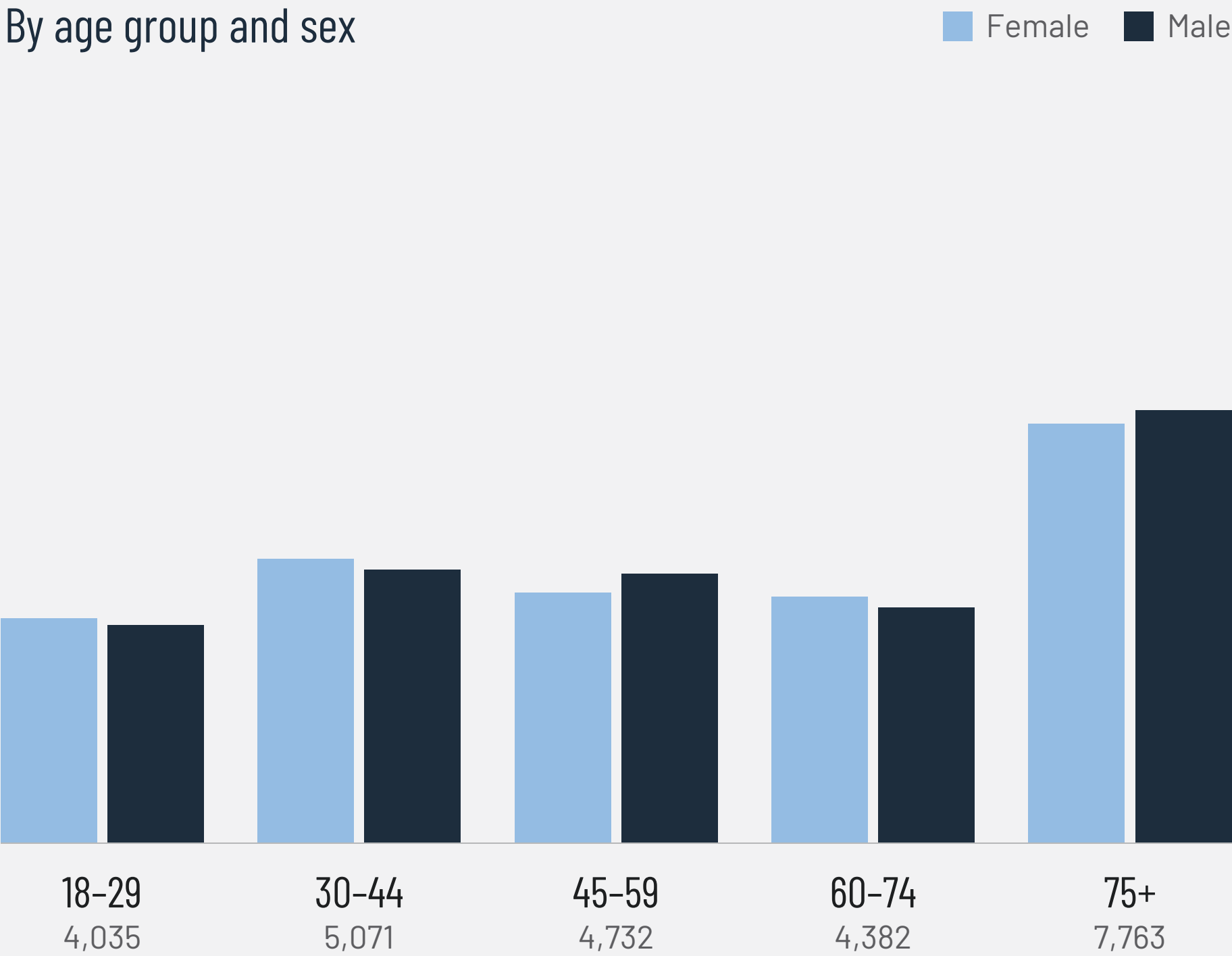
Central Station alone handles 52% of all vaccinations — the clear capacity bottleneck if demand grows.



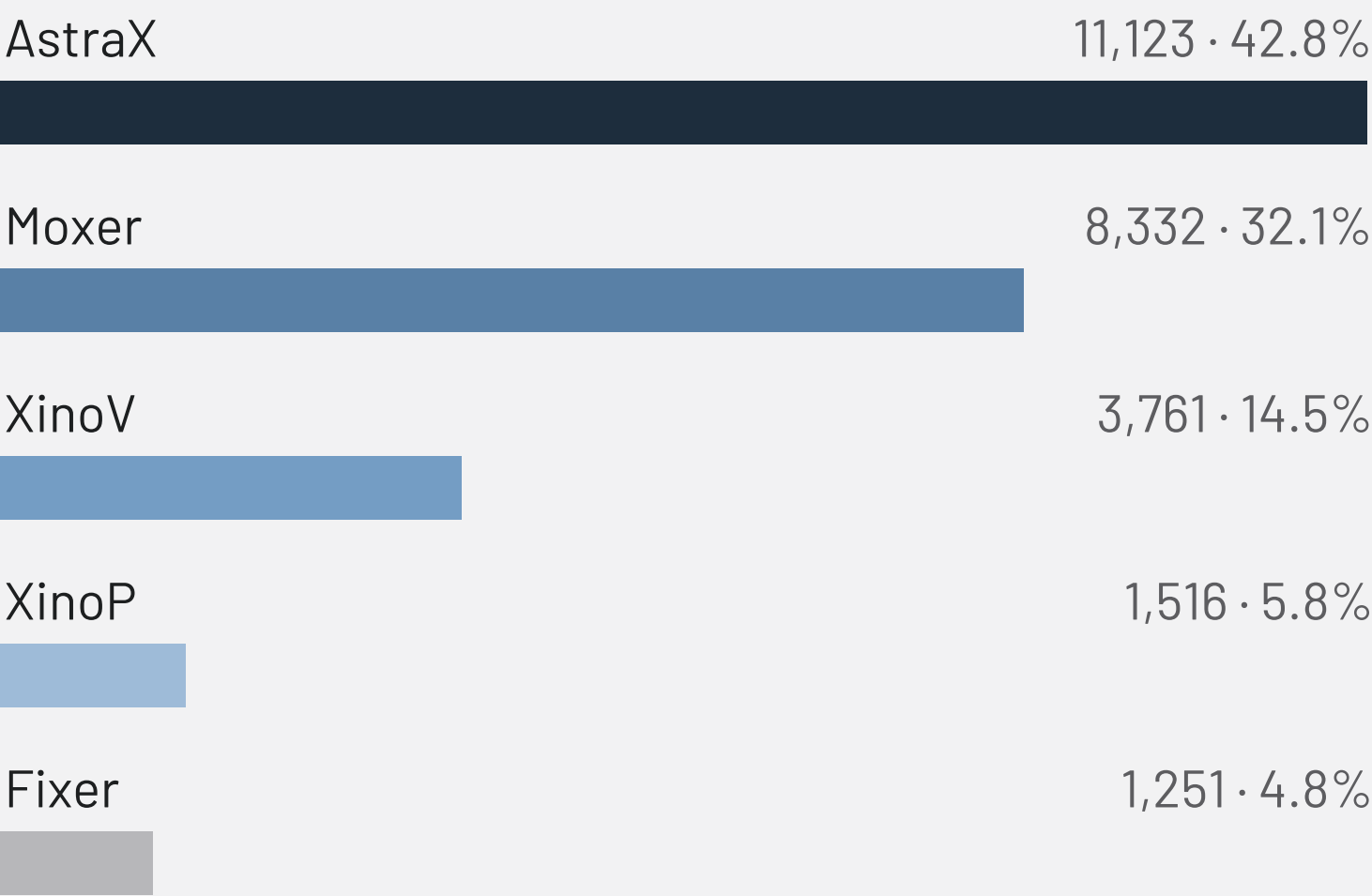
AGE SKEWS OLDER • TWO PRODUCTS DOMINATE

VOLUME BY POPULATION AND VACCINE

By age group and sex



By vaccine product

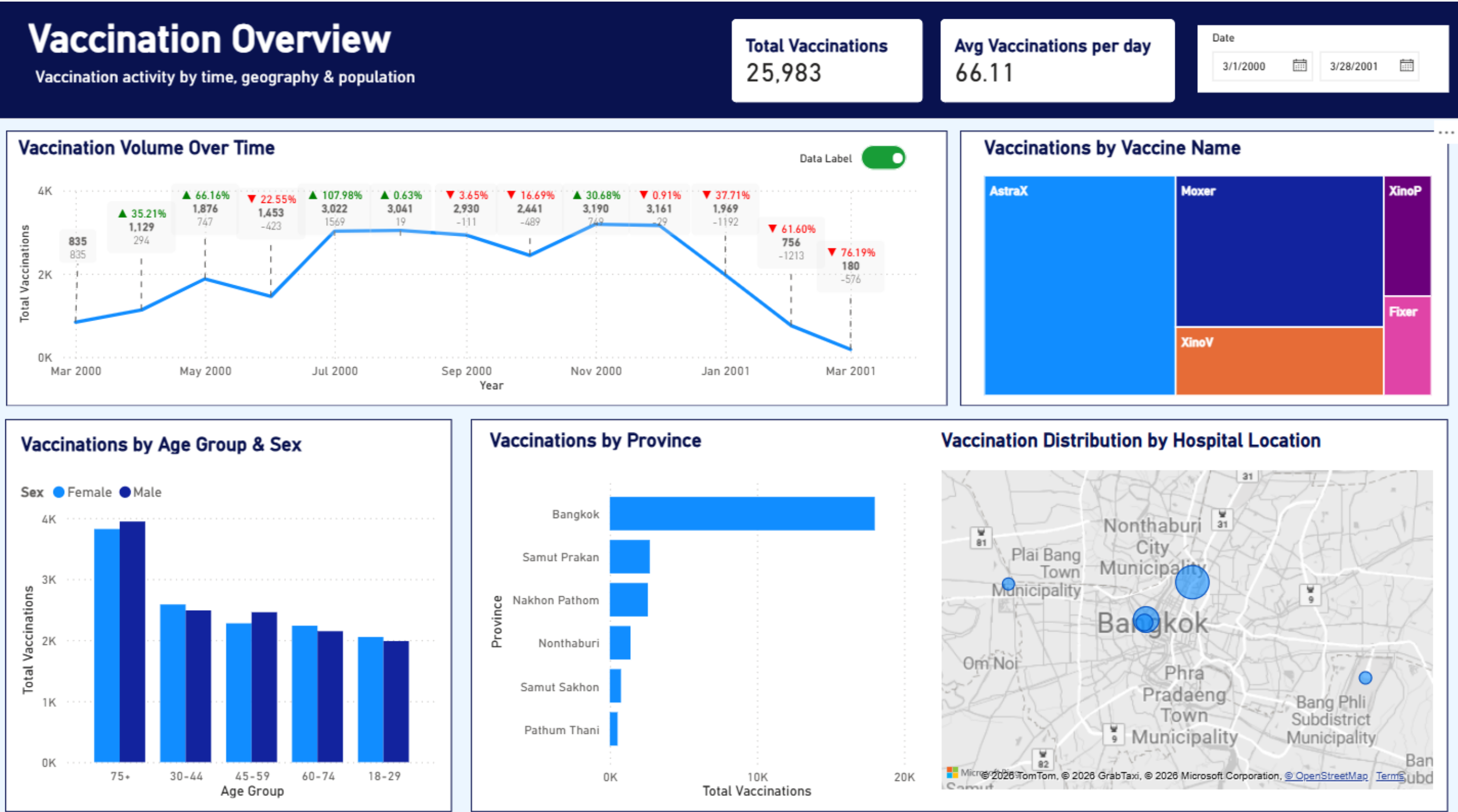


AstraX and Moxer together are 74.9% of doses administered.



INTERACTIVE FILTERS · ONE SHARED MEASURE

LIVE POWER BI DASHBOARD





INSIGHTS AND RECOMMENDATIONS

Demand is seasonal

Jul-Dec 2000 sustained ~2,960 doses a month, more than double the Mar-Apr start. Plan staffing and stock around this window, not a flat annual average.

Older adults are the largest group

Ages 75+ account for 30% of doses — the single largest age band, nearly double the 18-29 group. Outreach and scheduling should account for this skew.

One site is a bottleneck

Central Station delivers 52% of all doses. It should be first in line for added capacity and a natural candidate for load-balancing to nearby sites.

Fix the ID pipeline upstream

37 anonymised Person Ids collided by chance in a narrow ID space. Widen the anonymisation range so future extracts don't need a composite-key workaround.



SIDATA+ DATA ANALYTICS CANDIDATE ASSESSMENT

THANK YOU

Questions and discussion welcome.

Joy · Siriraj Informatics and Data Innovation Center